1 Limits of Trigonometric Functions

In this section we will find all limits of the six trig functions. First, we review the relevant inequalities and trig identities we will need.

The definitions of the sine and cosine functions imply the following inequalities for all $\theta \in \mathbb{R}$:

$$|\sin(\theta)| \leq 1, \tag{1.1}$$
$$|\cos(\theta)| \leq 1. \tag{1.2}$$

We will also need the following inequality, which is less obvious.

Proposition 1.3. *For all real numbers* θ*,*

$$|\sin(\theta)| \leq |\theta|. \tag{1.4}$$

Proof. If $|\theta| \geq 1$, then

$$\begin{aligned} |\sin(\theta)| &\leq 1 \quad \text{(by (1.1))} \\ &\leq |\theta| \quad \text{(since } 1 \leq |\theta| \text{ by assumption),} \end{aligned}$$

so the proposition is true. All that remains is to prove the proposition in the case where $|\theta| < 1$. Note also that if $\theta = 0$, then

$$|\sin(0)| = |0|,$$

so the proposition is true when $\theta = 0$.

First note that, for $0 < \theta < 1$, $\sin(\theta) > 0$, while for $-1 < \theta < 0$, $\sin(\theta) < 0$. Thus the inequality in (1.4) becomes

$$\sin(\theta) \leq \theta, \quad 0 < \theta < 1 \tag{1.5}$$

and

$$-\sin(\theta) \leq -\theta, \quad -1 < \theta < 0$$

or

$$\sin(\theta) \geq \theta, \quad -1 < \theta < 0. \tag{1.6}$$

Note that (1.5) and (1.6) are actually equivalent. To see this, note that if (1.5) holds, then multiplying (1.5) on both sides by -1 gives

$$-\sin(\theta) \geq -\theta, \quad 0 < \theta < 1$$

and, since $-\sin(\theta) = \sin(-\theta)$,

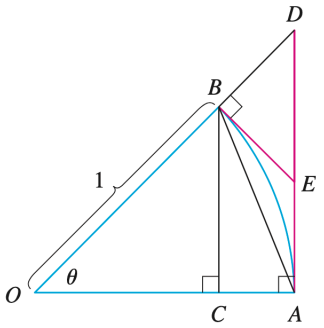
$$\sin(-\theta) \geq -\theta, \quad 0 < \theta < 1.$$

Defining $\phi = -\theta$, then (1.5) becomes

$$\sin(\phi) \geq \phi, \quad 0 < -\phi < 1 \iff 0 > \phi > -1 \iff -1 < \phi < 0. \tag{1.7}$$

If we change the variable name from ϕ back to θ, (1.7) is precisely (1.6). This proves that (1.5) implies (1.6). One similarly shows that (1.6) implies (1.5). Thus, to prove (1.4) for $0 < |\theta| < 1$, it is sufficient to prove (1.5).

To prove (1.5), consider the figure below which shows a sector of the unit circle with center O and central angle θ, where $0 < \theta < 1$.



Segment $\overline{BC}$ is perpendicular to $\overline{OA}$. By definition of radian measure, we have $m\widehat{AB} = \theta$. We also have $BC = OB\sin(\theta) = 1 \cdot \sin(\theta) = 1$. Since $\overline{AB}$ is the hypotenuse of the right triangle $\triangle ABC$, we have

$$BC < AB < m\widehat{AB}.$$

By substitution,

$$\sin(\theta) < AB < \theta,$$

which completes the proof. □

Finally, we will also need the trig identities:

$$\sin(\alpha) - \sin(\beta) = 2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right), \tag{1.8}$$

$$\cos(\alpha) - \cos(\beta) = -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right). \tag{1.9}$$

Theorem 1.10 (Limits of sine and cosine)**.** *For $a \in \mathbb{R}$,*

$$\lim_{x\to a}\sin(x) = \sin(a), \tag{1.11}$$

$$\lim_{x\to a}\cos(x) = \cos(a). \tag{1.12}$$

Proof. We prove the theorem for $\sin(x)$ and leave the proof for $\cos(x)$ as an exercise. Let $\varepsilon > 0$. Then by (1.8)

$$\begin{aligned}
|\sin(x) - \sin(a)| &= \left|2\cos\left(\frac{x+a}{2}\right)\sin\left(\frac{x-a}{2}\right)\right| \\
&= 2\left|\cos\left(\frac{x+a}{2}\right)\right|\left|\sin\left(\frac{x-a}{2}\right)\right| \\
&\le 2\left|\sin\left(\frac{x-a}{2}\right)\right| \qquad \text{(by (1.2))} \\
&\le 2\left|\frac{x-a}{2}\right| \qquad \text{(by (1.4))} \\
&= |x-a|
\end{aligned}$$

Thus, choose $\delta = \varepsilon$, so that by taking $|x-a| < \varepsilon$, we have $|\sin(x) - \sin(a)| < \varepsilon$. This proves that $\lim_{x\to a} \sin(x) = \sin(a)$. □

In particular, Theorem 1.10 shows that the sine and cosine functions have no infinite limits, hence their graphs have no vertical asymptotes. Intuitively, since the graphs of sine and cosine oscillate between 1 and -1, they do not approach a fixed number, so their limits at infinity do not exist. We now prove this rigorously.

Theorem 1.13 (Limits at infinity of sine and cosine do not exist)**.** *The limits at infinity* $\lim_{x\to\pm\infty} \sin(x)$ *and* $\lim_{x\to\pm\infty} \cos(x)$ *do not exist.*

Proof. Given any $\delta > 0$, we can find $x_1, x_2 > \delta$ such that $\sin(x_1) = 1$ and $\sin(x_2) = -1$.[1] We have therefore found two points on the portion of the graph of $\sin(x)$ to the right of $x = \delta$, such that the distance between them, $|\sin(x_1) - \sin(x_2)|$, is equal to 2. It is therefore not possible for all points on the graph of $\sin(x)$ to fit within an interval of size 2ε when $\varepsilon < 1$. This shows that $\lim_{x\to\infty} \sin(x) \neq L$ for any real number L. This also shows that $\lim_{x\to\infty} \sin(x) \neq \infty$; given $\varepsilon > 1$, for any $\delta > 0$ there exists $x_1 > \delta$ such that $\sin(x_1) = 1 < \varepsilon$. Similarly, $\lim_{x\to\infty} \sin(x) \neq -\infty$. Hence, $\lim_{x\to\infty} \sin(x)$ does not exist. □

It follows from Theorem 1.10 that limits of the other 4 trig functions at points in their domain may be evaluated by direct substitution.

Theorem 1.14 (Limits of trig functions at points in their domain)**.** *If* $f(x) = \tan(x), \cot(x), \sec(x),$ *or* $\csc(x)$ *and a a point in the domain of* f*, then*

$$\lim_{x\to a} f(x) = f(a).$$

Proof. We prove this for $\tan(x)$ and leave the remaining cases as an exercise. Suppose a is in the domain of $\tan(x)$. This is true if and only if $\cos(a) \neq 0$. Then, by the Quotient Law,

$$\begin{aligned}\lim_{x\to a} \tan(x) &= \lim_{x\to a} \frac{\sin(x)}{\cos(x)} \\ &= \frac{\lim_{x\to a} \sin(x)}{\lim_{x\to a} \cos(x)} \\ &= \frac{\sin(a)}{\cos(a)} \\ &= \tan(a).\end{aligned}$$

□

One can prove that $\tan(x), \cot(x), \sec(x)$, or $\csc(x)$ has a vertical asymptote at each point which is not in the domain.

Theorem 1.15 (Limits of trig functions at points not in their domain)**.** *The trig functions* $\tan(x), \cot(x), \sec(x),$ *and* $\csc(x)$ *have a vertical asymptote at each point that is not in their domain.*

[1] Explicitly, given $\delta > 0$, we can find $n \in \mathbb{N}$ such that $n > \delta$. Then $2n > \delta$, so without loss of generality we may assume that n is even. Note that this implies $n+1$ is odd. Then $(n+1+\frac{1}{2})\pi > (n+\frac{1}{2})\pi > \delta$ and $\sin((n+\frac{1}{2})\pi) = 1$, while $\sin((n+1+\frac{1}{2})\pi) = -1$.

The proof of this theorem in full generality would be unbearably tedious. To give an idea of how this is done, we prove that $\lim_{x\to\frac{\pi}{2}} \sec(x) = -\infty$ below. In practice, to remember which one-sided limits are $+\infty$ and which are $-\infty$, it is best to just memorize the graphs of these functions.

Proof. To prove that $\lim_{x\to\frac{\pi}{2}} \sec(x) = -\infty$, we need to show that, given $\varepsilon > 0$, there exists $\delta > 0$ such that $0 < |x - \frac{\pi}{2}| < \varepsilon$ implies $\sec(x) < -\varepsilon$. By the trig identity (1.9) with $\alpha = x$ and $\beta = \frac{\pi}{2}$, we have

$$\begin{aligned}\sec(x) &= \frac{1}{\cos(x)} \\ &= \frac{1}{-2\sin\left(\frac{x+\frac{\pi}{2}}{2}\right)\sin\left(\frac{x-\frac{\pi}{2}}{2}\right)}\end{aligned}$$

Suppose $0 < x - \frac{\pi}{2} < \frac{\pi}{2}$. Then $0 < \frac{x-\frac{\pi}{2}}{2} < \frac{\pi}{4}$ implies

$$1 \geq \sin\left(\frac{x-\frac{\pi}{2}}{2}\right) > 0$$

and

$$\begin{aligned}0 < x - \frac{\pi}{2} &< \frac{\pi}{2} \\ \pi < x + \frac{\pi}{2} &< \frac{3\pi}{2} \\ \frac{\pi}{2} < \frac{x+\frac{\pi}{2}}{2} &< \frac{3\pi}{4}\end{aligned}$$

which implies

$$1 \geq \sin\left(\frac{x+\frac{\pi}{2}}{2}\right) > 0$$

and therefore

$$\frac{1}{\sin\left(\frac{x+\frac{\pi}{2}}{2}\right)} \geq 1$$

from which it follows that

$$\begin{aligned}-\frac{1}{2\sin\left(\frac{x+\frac{\pi}{2}}{2}\right)\sin\left(\frac{x-\frac{\pi}{2}}{2}\right)} &\leq -\frac{1}{2\sin\left(\frac{x-\frac{\pi}{2}}{2}\right)} \\ &\leq -\frac{1}{2\left(\frac{x-\frac{\pi}{2}}{2}\right)} \\ &= -\frac{1}{x-\frac{\pi}{2}},\end{aligned}$$

where we have used the fact that if $0 < \alpha < \frac{\pi}{4}$, then we have $0 < \sin(\alpha) < |\sin(\alpha)|$ and $0 < \alpha = |\alpha|$, so (1.4) becomes $\sin(\alpha) \leq \alpha$. It follows that, by taking $\delta = \min\{\frac{\pi}{2}, \frac{1}{\varepsilon}\}$, then $0 < x - \frac{\pi}{2} < \delta$ implies

$$\sec(x) \leq -\frac{1}{x - \frac{\pi}{2}} < -\varepsilon.$$

□

Similarly, one can prove that $\tan(x), \cot(x), \sec(x)$, and $\csc(x)$ have no limits at infinity. The proof is again unbearably tedious, and in practice one should just memorize the graphs of these functions.

Theorem 1.16 (Trig functions have no limits at infinity)**.** *The trig functions* $\tan(x), \cot(x), \sec(x)$*, and* $\csc(x)$ *have no limits at infinity.*

2 Limits of exponential functions

Theorem 2.1 (Limits of exponential functions may be evaluated by direct substitution)**.** *Let $b > 0$. Then for $a \in \mathbb{R}$,*

$$\lim_{x \to a} b^x = b^a.$$

Proof. The cases $0 < b < 1$, $b = 1$, and $b > 1$ need to be analyzed separately. We will do the latter two, and leave the former as an exercise. If $b = 1$, then $\lim_{x\to a} b^x = \lim_{x\to a} 1^x = \lim_{x\to a} 1 = 1 = 1^a$, so the claim is true.

Suppose $b > 0$. Then b^x is strictly increasing for all real numbers. We now compute both one-sided limits. To prove that $\lim_{x\to a^+} b^x = b^a$ we need to show that, given $\varepsilon > 0$ there exists $\delta > 0$ such that $|b^x - b^a| < \varepsilon$ whenever $0 < |x - a| < \delta$. Given $\varepsilon > 0$, choose $\delta = \log_b\left(\frac{\varepsilon}{b^a} + 1\right)$. Note that, since $\varepsilon > 0$ and $b^a > 0$, $\frac{\varepsilon}{b^a} + 1 > 1$ and therefore $\delta > 0$. If $0 < x - a < \delta$ then since b^x is strictly increasing, we have

$$b^0 = 1 < b^{x-a} < b^\delta = b^{\log_b(\frac{\varepsilon}{b^a} + 1)} = \frac{\varepsilon}{b^a} + 1.$$

It follows that

$$\begin{aligned} |b^x - b^a| &= |b^a(b^{x-a} - 1)| \\ &= |b^a||b^{x-a} - 1| \\ &= b^a|b^{x-a} - 1| \\ &< b^a|\frac{\varepsilon}{b^a} + 1 - 1| \\ &= b^a|\frac{\varepsilon}{b^a}| \\ &= b^a \cdot \frac{\varepsilon}{b^a} \\ &= \varepsilon. \end{aligned}$$

This proves that $\lim_{x\to a^+} b^x = b^a$.

To prove that $\lim_{x\to a^-} b^x = b^a$, we need to show that, given $\varepsilon > 0$ there exists a corresponding $\delta > 0$ such that $0 < a - x < \delta$ implies $|b^x - b^a| < \varepsilon$. Without loss of generality, suppose we are given ε such that $0 < \varepsilon < b^a$. [2] Corresponding to this ε, choose $\delta = -\log_b(1 - \frac{\varepsilon}{b^a})$. Note that $0 < 1 - \frac{\varepsilon}{b^a} < 1$, so the log is defined and $\delta > 0$. If $0 < a - x < \delta$, then since b^x is increasing we have

$$\begin{aligned} &0 < a - x < \delta \\ &0 > x - a > -\delta \\ &b^{x-a} > b^{-\delta} = b^{\log_b(1-\frac{\varepsilon}{b^a})} = 1 - \frac{\varepsilon}{b^a} \end{aligned}$$

[2] There is no loss of generality here, since if there exists $\delta > 0$ such that $0 < a - x < \delta$ implies $|b^x - b^a| < \varepsilon < b^a$, then we will have $|b^x - b^a| < \varepsilon$ for any larger ε as well.

and therefore

$$\begin{aligned} b^{x-a}-1 &> -\frac{\varepsilon}{b^a} \\ -(b^{x-a}-1) &< \frac{\varepsilon}{b^a} \quad \text{(since } b^{x-a}<1 \text{ when } x<a\text{)} \\ |b^{x-a}-1| &< \frac{\varepsilon}{b^a} \\ b^a|b^{x-a}-1| &< \varepsilon \\ |b^x-b^a| &< \varepsilon. \end{aligned}$$

This proves that $\lim_{x\to a^-} b^x = b^a = \lim_{x\to a^+} b^x$, and therefore proves that $\lim_{x\to a} b^x = b^a$ when $b>1$. □

Corollary 2.2. *For $a\in\mathbb{R}$, $\lim_{x\to a} e^x = e^a$.*

The previous theorem implies that exponential functions have no infinite limits.

Theorem 2.3 (Limits at infinity of exponential functions)**.** *Let $b>0$.*

(1) If $b>1$, $\lim_{x\to-\infty} b^x = 0$ and $\lim_{x\to\infty} b^x = \infty$.

(2) If $b=1$, then $\lim_{x\to-\infty} b^x = \lim_{x\to\infty} b^x = 0$.

(3) If $b>1$, $\lim_{x\to-\infty} b^x = \infty$ and $\lim_{x\to\infty} b^x = 0$.

Proof. Case (2) is obvious, since $1^x = 1$ is a constant function.

Suppose $b>0$. Let $\varepsilon>0$. Without loss of generality, suppose $\varepsilon>1$. [3] choose $\delta = \log_b(\varepsilon)$. Since $\varepsilon>1$, it follows that $\delta>0$. If $x>\delta$ then since b^x is increasing when $b>1$, it follows that $b^x > b^\delta = b^{\log_b(\varepsilon)} = \varepsilon$.

The remaining cases are left as an exercise. □

Corollary 2.4. *$\lim_{x\to-\infty} e^x = 0$ and $\lim_{x\to\infty} e^x = \infty$.*

Again, all the limits at infinity above are best remembered by memorizing the graph of b^x for the various cases.

3 Summary

- Limits of trig functions at points in their domain can be evaluated by direct substitution.
- Limits of exponential functions can be evaluated by direct substitution.

[3] There is no loss of generality here in taking $\varepsilon>1$, since we are trying to show that we can make b^x arbitrarily large.